Card Fraud Triage

Gaussian mixture density scoring

Project 26
Expanded machine learning evaluation

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Fraud triage and evaluation scope

492 frauds / 284,807 transactions

Training learns legitimate behavior. Scores rank review priority.

AP and review workload measure rare-event triage quality.

The expanded run reuses a previously inspected test partition.

Autoencoder training remains Phase 2.

Frozen chronological membership

Partition	All records	Frauds	Fraud rate
train	170,888	360	0.21066%
validation	56,958	57	0.10007%
test	56,961	75	0.13167%

170,528 legitimate training records for every primary model.
The same saved row IDs are reused for Phase 2.

Preprocessing and model inputs

29 inputs: V1–V28 and log(1 + Amount)

StandardScaler fits only legitimate training observations.

Time defines partitions and EDA. Class never enters X.

PCA sizes 5, 10 and 20 support the dimensionality study.

Saved raw-frame pipelines preserve transformations and order.

Gaussian mixture density scoring

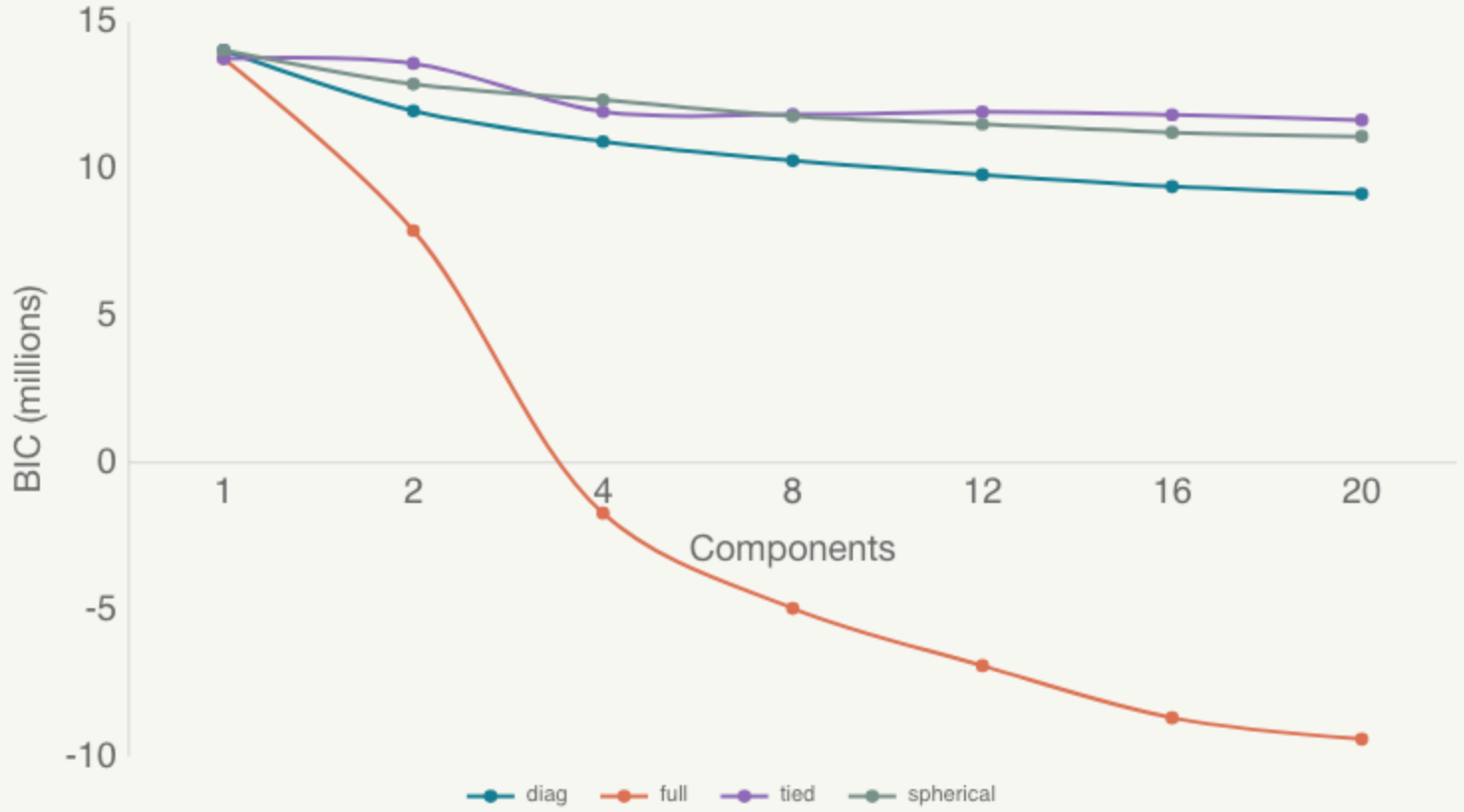
$$p(x) = \Sigma\, \pi_k\, N(x\,;\, \mu_k, \Sigma_k)$$

Anomaly score = $-\log p(x)$

EM alternates responsibilities with weighted parameter updates.

Regularization, restarts and iteration limits affect fitting.
Convergence and conditioning require separate checks.

Train-only structural selection



Selected
20 components
full covariance

reg_covar: 0.000001
n_init: 2
max_iter: 300

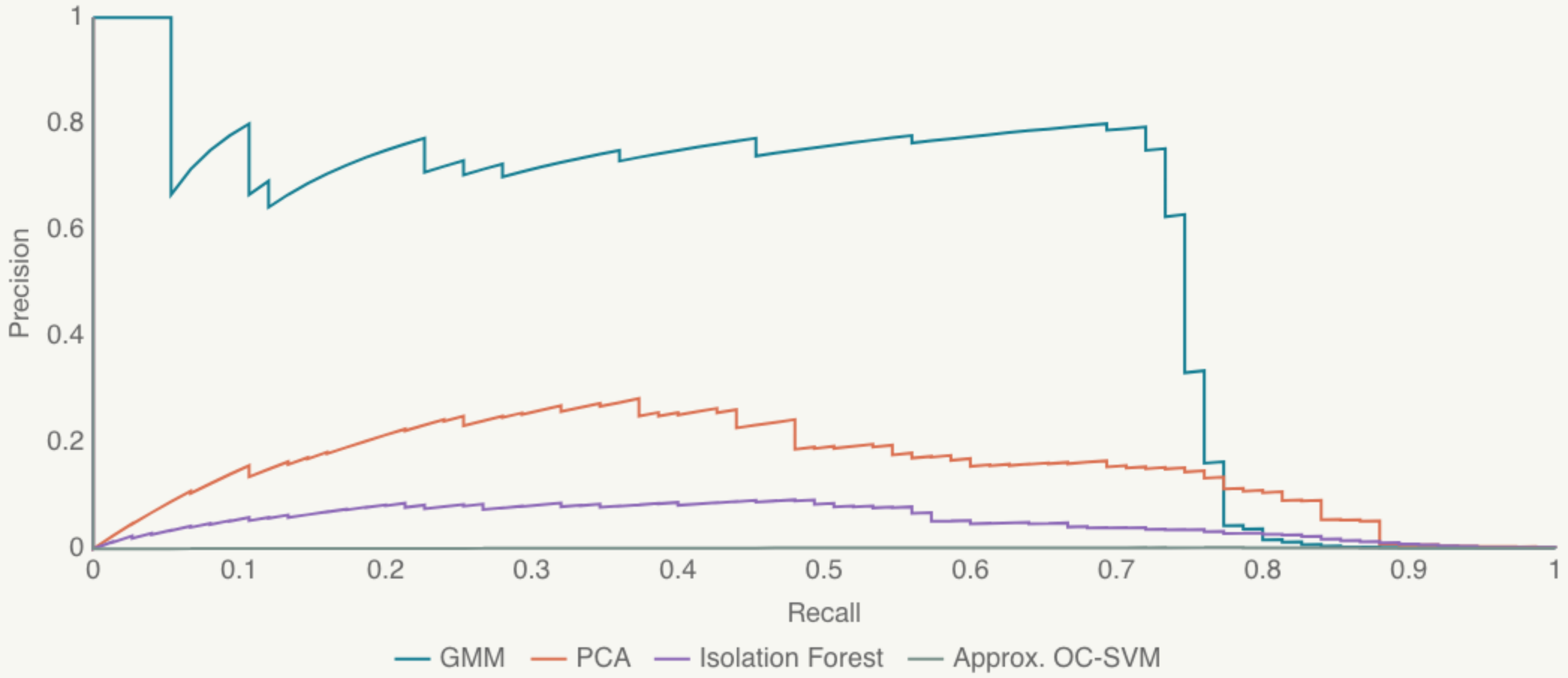
Bounded staged search

Classical comparators

Model	Configuration	Anomaly score
GMM	20 full covariance	Negative log density
Isolation Forest	200 trees, 256 samples/tree	Negative normality
PCA	15 components, full SVD	Reconstruction MSE
Approx. OC-SVM	128 RBF features + SGD	Negative normality

Weak configurations remain in the comparison.
These findings do not establish failure of an entire model family.

Reused-test precision and recall



Ranking results and uncertainty

Model	Average precision	95% AP interval	PR AUC
GMM	0.5822	0.4634–0.7123	0.5787
Isolation Forest	0.0538	0.0399–0.0718	0.0524
PCA	0.1573	0.1212–0.2080	0.1540
Approx. OC-SVM	0.0014	0.0013–0.0016	0.0014

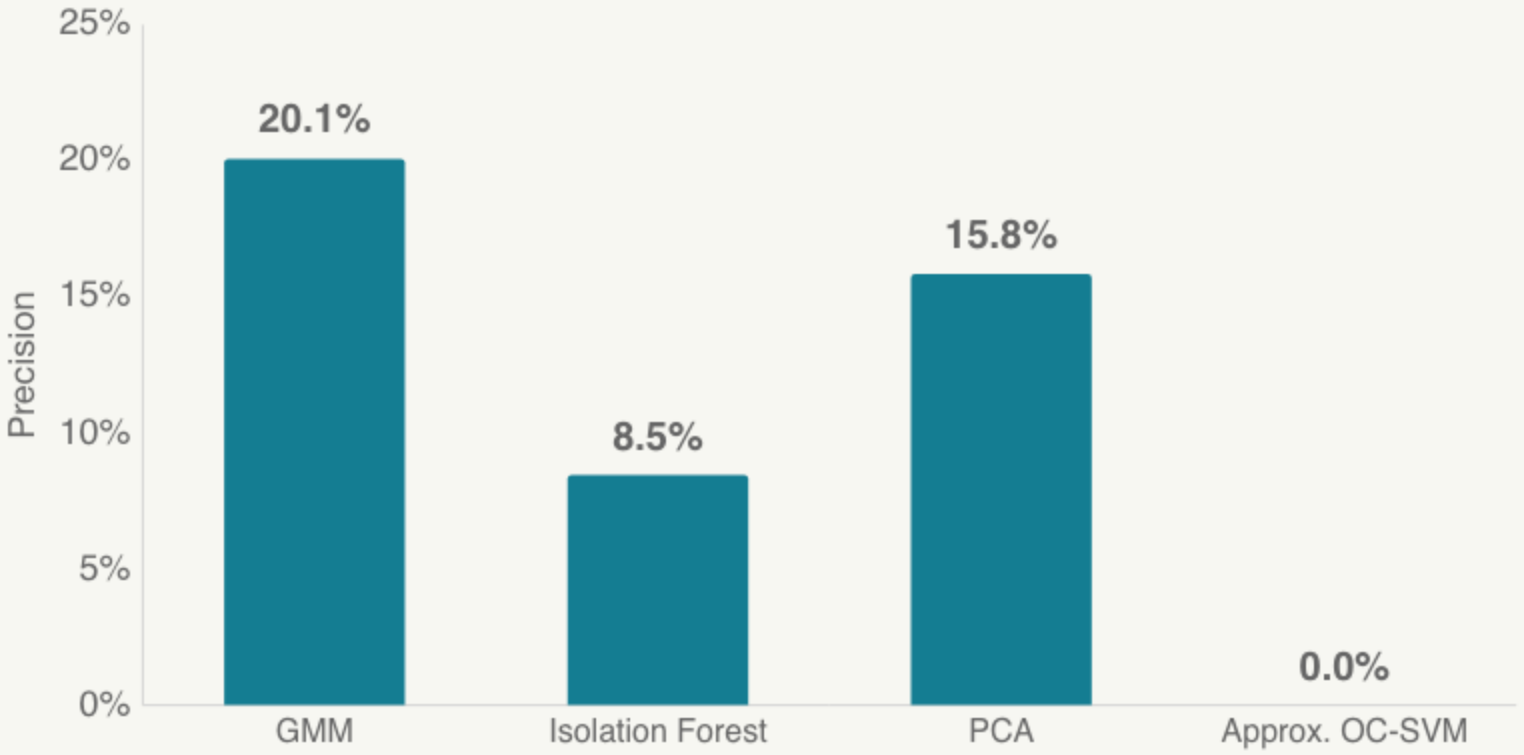
Training BIC chooses the GMM.
Test AP reports outcomes and does not drive retuning.

Frozen thresholds and confusion counts

Validation budget	Test TP	Test FP	Test recall	Actual test FPR
0.1%	55	29	73.33%	0.051%
0.5%	57	175	76.00%	0.308%
1.0%	58	324	77.33%	0.570%
5.0%	60	2082	80.00%	3.660%

At the 0.5% budget: 232 alerts, 57 frauds, 175 false alarms.
3.07 legitimate alerts per caught fraud.

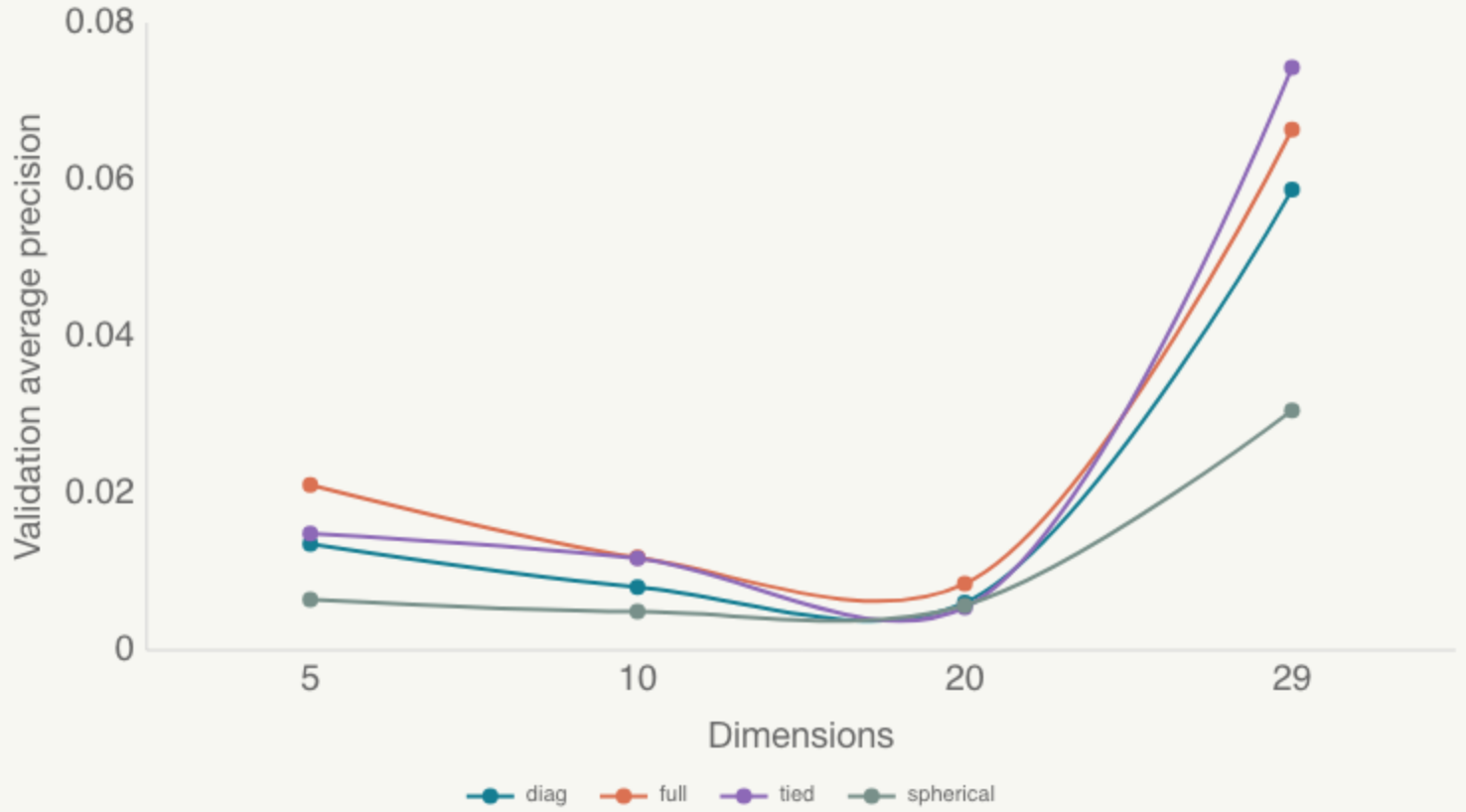
Precision at a fixed review capacity



284 reviews
GMM catches
57 / 75 frauds

Top 100 catches
56 / 75 frauds

Validation-only dimensionality study



26 diagnostic fits

Dimension changes both information and conditioning.

No test-based tuning

Seed and sample-size sensitivity

Covariance	Seeds	Mean validation AP	Sample SD
diag	3	0.0589	0.0002
full	3	0.3534	0.2488
spherical	3	0.0298	0.0016
tied	3	0.1075	0.0581

Nested training subsets: 10%, 50%, 100%.
Conditioning, warnings and score-rank correlations accompany AP.

Post-freeze error analysis

Group	Records	Median amount	Median score
caught_fraud	57	2.00	155.02
false_positive	175	157.43	71.37
missed_fraud	18	7.79	-20.50
true_negative	56711	17.99	-42.66

Inspect rare legitimate behavior and normal-looking fraud.
Anonymized V features limit business interpretation.

Limits and Phase 2 freeze

Historical coverage: two days and 75 test frauds.

The test set is reused. The search is bounded and staged.

Three seeds and fixed K=4 limit robustness conclusions.

Freeze split IDs, preprocessing and the evaluation harness.

Phase 2 needs validation choices and justified hybrid ablations.